

Classifying Regular Lagrangian Fibrations over Compact Surfaces

Michael McGloin (Luka Zwaan)

KU Leuven

May 6, 2026

Overview

- ① (Integral) Affine Manifolds
- ② Symplectic geometry and Lagrangian fibrations
- ③ Classifying Integral Affine Structures
- ④ Classification of Lagrangian Fibrations

Section 1

(Integral) Affine Manifolds

Affine Manifolds

Definition

An *affine manifold* is a differentiable manifold such that the transition maps lie in the affine group $\text{Aff}(\mathbb{R}^n)$

Remark

The group $\text{Aff}(\mathbb{R}^n) := GL(\mathbb{R}^n) \ltimes \mathbb{R}^n$. An element $g \in \text{Aff}(\mathbb{R}^n)$ acts on $x \in \mathbb{R}^n$ by $g(x) = Ax + b$ where $A \in GL(\mathbb{R}^n)$ and $b \in \mathbb{R}^n$.

(Integral) Affine Manifolds

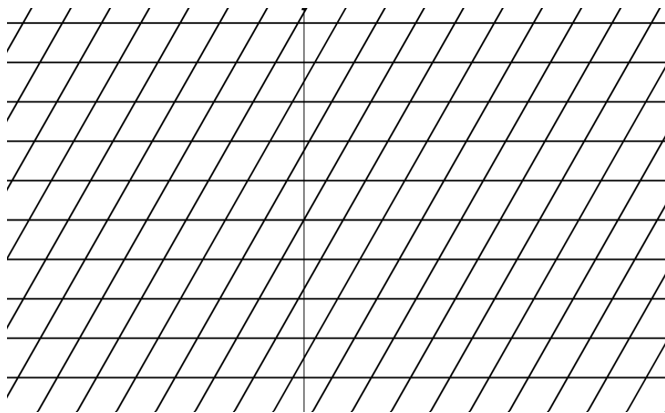


Figure: Affine shear transformation

The Affine Connection

Proposition

An affine manifold M comes equipped with a flat, torsion free connection ∇ . In affine coordinates this appears as the standard Euclidean connection.

Remark

Conversely, a manifold M equipped with a flat, torsion free connection can be endowed with an affine structure.

The Development Pair

Lemma (Ehresmann, 1936)

Let M be an affine manifold and denote by $\tilde{M}$ its universal cover. Then there exists a pair (dev, hol) called the *development pair* where

$$\text{dev} : \tilde{M} \rightarrow \mathbb{R}^n,$$

$$\text{hol} : \pi_1(M) \rightarrow \text{Aff}(\mathbb{R}^n)$$

The linear part of hol is the holonomy of the affine connection ∇ .

Complete Affine Manifolds

Definition

An affine manifold is said to be complete if the development map is a global diffeomorphism. This is equivalent to completeness of the connection ∇ .

Remark

If a discrete group G acts freely and properly on $\mathbb{R}^n$ by affine transformations then $\mathbb{R}^n/G$ will inherit a complete affine structure.

Integral Affine Manifolds

Definition

An *integral affine manifold* is an affine manifold such that the transition maps lie in the integral affine group $\text{Aff}_{\mathbb{Z}}(\mathbb{R}^n)$ - e.g. the linear part is in $GL(\mathbb{Z}^n)$

Remark

An matrix $A \in GL(\mathbb{Z}^n)$ has $\det A = \pm 1$.

Integral Affine Manifolds

Lemma

An integral affine structure on a manifold M^n is equivalent to a rank- n lattice in T^*M locally spanned by closed 1-forms

Examples of Integral Affine Manifolds

- ▶ $\mathbb{R}^n$ with the standard integral affine structure.
- ▶ The torus $(\mathbb{R}/\mathbb{Z})^n$ with the integral affine structure induced by translational action of $\mathbb{Z}^n$.
- ▶ Let $(M, \mathcal{A})$ be an integral affine manifold and suppose that

$$f : N \rightarrow M$$

is a local diffeomorphism. Then we can pullback the integral affine structure from M to N .

Section 2

Symplectic geometry and Lagrangian fibrations

Lagrangian Submanifold

Definition (Lagrangian Submanifold)

Let (M, ω) be a symplectic manifold and let $N \subset M$ be a submanifold. N is said to be a *Lagrangian submanifold* of M if $\dim N = \frac{1}{2} \dim M$ and $\omega|_N = 0$.

Example

The cotangent bundle (T^*B, ω_{can}) has a canonical symplectic structure. The fibres of $\pi : T^*B \rightarrow B$ are Lagrangian submanifolds of T^*B .

Lagrangian Fibration

Definition

A Lagrangian fibration $\pi : (M, \omega) \rightarrow B$ is a surjective map whose (regular) fibres are Lagrangian submanifolds of (M, ω) .

Remark

Throughout this talk we will assume that π is a surjective submersion and a proper map so that by a theorem of Ehresmann, 1951, $\pi : (M, \omega) \rightarrow B$ is a fiber bundle.

Examples of Lagrangian Fibrations

- ▶ $(T^*B, \omega_{can}) \rightarrow B$ is a Lagrangian fibration.
- ▶ $(T^*B, \omega_{can} + \pi^*\beta) \rightarrow B$ - where $\beta \in \Omega^2(B)$ is closed.
- ▶ Completely integrable Hamiltonian systems. (M^{2n}, ω) and $F = f_1, \dots, f_n : M \rightarrow \mathbb{R}^n$ with

$$\{f_i, f_j\} = 0$$

$$df_1 \wedge \dots \wedge df_n \neq 0$$

almost everywhere.

Integral Affine structure on the Base

Let $\rho : M \rightarrow B$ be a Lagrangian fibration.

- ▶ $T_b^*B \curvearrowright M_b$ fibrewise. Given $\alpha \in \Omega^1(B)$ we associate $X_\alpha \in \mathfrak{X}(M)$ by

$$\iota_{X_\alpha} \omega = -\rho^* \alpha.$$

- ▶ Denoting the time 1 flow of X_α by $F_\alpha : M \rightarrow M$ this is a fibrewise transitive, vector bundle action of T^*B on M .

Integral Affine structure on the Base

Let $\Lambda \subset T^*B$ be the bundle of stabilisers for this action.

Lemma

Λ is a discrete subbundle of T^*B locally spanned by closed one-forms. In fact, the symplectic form ω_{can} descends to the quotient and $\pi : T^*B/\Lambda \rightarrow B$ is a Lagrangian fibration.

Remark

*The lattice $\Lambda \subset T^*B$ gives B the structure of an integral affine manifold.*

Integral Affine structure on the Base

Definition

The Lagrangian fibration $\pi : T^*B/\Lambda \rightarrow B$ will be called the symplectic reference Lagrangian fibration with period lattice Λ .

Integral Affine structure on the Base

Lemma

Let $M \rightarrow B$ and $M' \rightarrow B$ be Lagrangian fibrations. Then if there is a fibrewise symplectomorphism covering the identity $\varphi : M \rightarrow M'$ then the induced integral affine structure on B is the same for both fibrations.

Remark

*The converse for this does not hold. Adding a non-exact, closed 2-form to (T^*B, ω_{can}) is an example.*

Section 3

Classifying Integral Affine Structures

Classifying Complete Integral Affine Structures

- ▶ Completeness forces injectivity of holonomy representation
- ▶ Classifying complete structures $\longleftrightarrow$ classifying injective holonomy representations inducing free and proper action on $\mathbb{R}^n$.

Possible Compact Affine Surfaces

Theorem (Milnor)

If M is a compact, orientable surface of genus $g \geq 2$, then it does not possess a flat affine connection.

Corollary

The only possible orientable compact surfaces admitting an (integral) affine structure are sphere S^2 and the torus T^2

Possible Compact Affine Surfaces

Lemma

S^2 does not admit an (integral) affine structure

Corollary

The only compact integral affine surfaces are the torus and Klein bottle.

Nilpotent Holonomy

Theorem (Fried et al., 1981)

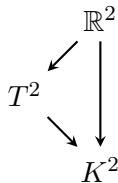
Let M be an orientable, compact affine manifold with nilpotent holonomy group. Then M is complete if and only if it possesses a parallel volume form.

Corollary

A compact, orientable, integral affine surface with nilpotent holonomy is complete.

Nilpotent Holonomy

- ▶ $\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}$
- ▶ $\pi_1(K^2) = \langle a, b \mid aba = b \rangle$



Nilpotent Holonomy

Corollary

Any integral affine structure on the 2-torus or the Klein bottle is complete.

Classification of Integral Affine Structures on $\mathbb{T}^2$ - $\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}$

Theorem (Mishachev, 1996)

The possible generators for the affine holonomy group are

- 1 $(I, \begin{pmatrix} a \\ c \end{pmatrix})$ and $(I, \begin{pmatrix} b \\ d \end{pmatrix})$ with $\det(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) \neq 0$ with $a, b, c, d \in \mathbb{R}$
- 2 $(I, \begin{pmatrix} a \\ 0 \end{pmatrix})$ and $(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ b \end{pmatrix})$ with $n \in \mathbb{N}$ and $a, b > 0$

Classification of Integral Affine Structures on $\mathbb{K}^2$ - $\pi_1(K^2) = \langle a, b \mid aba = b \rangle$

Let $A_1 = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$, $A_2 = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ and $B_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $B_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $B_3 = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$

Theorem (Sepe, 2010)

The possible generators for the affine holonomy group are

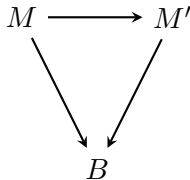
- 1 $(I, x\mathbf{e}_2)$ and $(B_1, y\mathbf{e}_1)$ with $x, y > 0$.
- 2 $(A_1, x\mathbf{e}_2)$ and $(B_1, y\mathbf{e}_1 + x\mathbf{e}_2)$ with $x, y > 0$.
- 3 $(I, x(\mathbf{e}_1 - \mathbf{e}_2))$ and $(B_2, y\mathbf{e}_2)$ with $x, y > 0$.
- 4 $(A_2, x\mathbf{e}_2)$ and $(B_3, y\mathbf{e}_1 + \frac{n-1}{n}x\mathbf{e}_2)$ with $2y \neq \frac{n-1}{n}x$ and $x > 0$.

Section 4

Classification of Lagrangian Fibrations

Goal

- Classify Lagrangian fibrations up to fibrewise symplectomorphism covering the identity.



How to classify the Lagrangian fibrations over a closed surface

- ▶ Classify the integral affine structures on B . (done)
- ▶ Topological classification - given by *Chern Class*
- ▶ Symplectic classification - given by *Lagrangian Class*

Luttinger Surgery

- ▶ Luttinger surgery allows us to build new Lagrangian fibrations from $T^*B/\Lambda \rightarrow B$.
- ▶ A type of "symplectic" Dehn surgery.
- ▶ Using this surgery it is possible to produce all Lagrangian fibrations $M \rightarrow B$ by:
 - Performing Luttinger surgery on T^*B/Λ .
 - Adding a cohomology class to the base.

The Chern Class

This will give the obstruction to $M \rightarrow B$ admitting a global section.

- ▶ Let $\{U_i\}_{i \in \mathbb{N}}$ be a trivialisation of $M \rightarrow B$.
- ▶ Pick sections s_i over U_i of $M \rightarrow B$.
- ▶ For some $\mu_{ij} : U_i \cap U_j \rightarrow T^*B/\Lambda$ we have $s_i = \mu_{ij} + s_j$. This defines a one-cocycle and hence an element $\mu \in H^1(B, T^*B/\Lambda)$ not depending on the choice of s_i .

The Chern Class

- ▶ There is a short exact sequence of sheaves

$$0 \rightarrow \Lambda \rightarrow \mathcal{T}^*B \rightarrow \mathcal{T}^*B/\Lambda \rightarrow 0$$

which leads to a long exact sequence in sheaf cohomology.

- ▶ $\mathcal{T}^*B$ is a fine sheaf $\implies H^i(\mathcal{T}^*B) = 0$ for $i > 0$.
- ▶ This gives an isomorphism $\delta : H^1(B, \mathcal{T}^*B/\Lambda) \rightarrow H^2(B, \Lambda)$

The Chern Class

Recall:

- 1 $s_i = \mu_{ij} + s_j$ defines a one-cocycle and hence an element $\mu \in H^1(B, \mathcal{T}^*B/\Lambda)$
- 2 $\delta : H^1(B, \mathcal{T}^*B/\Lambda) \rightarrow H^2(B, \Lambda)$ is an isomorphism.

Lemma (Duistermaat, 1980)

The image of μ under the isomorphism δ defines a class $[\mu_C] \in H^2(B, \Lambda)$. This class is the obstruction to our bundle $M \rightarrow B$ being topologically equivalent to the symplectic reference bundle $T^*B/\Lambda \rightarrow B$.

The Lagrangian Class

- ▶ If we require the sections of $\mu_{ij} := s_i - s_j$ to be Lagrangian we define a cohomology class $\hat{\mu}_L \in H^1(\mathcal{Z}(\mathcal{T}^*B/\Lambda))$
- ▶ There is a short exact sequence of sheaves

$$0 \rightarrow \Lambda \rightarrow \mathcal{Z}(\mathcal{T}^*B) \rightarrow \mathcal{Z}(\mathcal{T}^*B/\Lambda) \rightarrow 0$$

which leads to a long exact sequence in sheaf cohomology.

$$\cdots \rightarrow H^1(\Lambda) \rightarrow H^1(\mathcal{Z}(\mathcal{T}^*B)) \rightarrow H^1(\mathcal{Z}(\mathcal{T}^*B/\Lambda)) \xrightarrow{\Delta} H^2(\Lambda) \xrightarrow{\hat{d}} H^2(\mathcal{Z}(\mathcal{T}^*B)) \rightarrow \cdots$$

Relating the classes

- ▶ We have that $\mu_L \xrightarrow{\Delta} \mu_C \xrightarrow{\hat{d}} 0$.
- ▶ A necessary and sufficient condition for an element μ in $H^2(\Lambda)$ to be the Chern class of some Lagrangian torus fibration is that $\hat{d}(\mu) = 0$
- ▶ Every class $[\mu] \in H^1(\mathcal{Z}(T^*B/\Lambda))$ can be realized as the Lagrangian class of some Lagrangian torus fibration. (Luttinger surgery).
- ▶ Fixing μ_C such that $\hat{d}(\mu) = 0$, the elements μ_L such that $\Delta(\mu_L) = \mu_C$ are classified by a quotient of $H^2(B, \mathbb{R})$

The Torus Classification (Mishachev, 1996)

The Lagrangian fibrations $M \rightarrow \mathbb{T}^2$ are classified by

- ▶ The integral affine structure on the base.
 - $\mathcal{A}_{a,b,c,d} = \left\langle (I, \begin{pmatrix} a \\ c \end{pmatrix}), (I, \begin{pmatrix} b \\ d \end{pmatrix}) \right\rangle$.
 - $\mathcal{B}_{n,a,b} = \left\langle (I, \begin{pmatrix} a \\ 0 \end{pmatrix}), (\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 \\ b \end{pmatrix}) \right\rangle$
- ▶ An element $[s] \in H^2(\Lambda)$.
 - For $\mathcal{A}_{a,b,c,d}$ we have that $H^2(\Lambda) = \mathbb{Z} \oplus \mathbb{Z}$.
 - For $\mathcal{B}_{n,a,b}$ we have that $H^2(\Lambda) = \frac{\mathbb{Z}}{n\mathbb{Z}} \oplus \mathbb{Z}$
- ▶ An element $[h] \in H^2(\mathbb{T}^2, \mathbb{R})/N$ where N is some normal subgroup.

The Klein Bottle Classification (Sepe, 2010)

The Lagrangian fibrations $M \rightarrow \mathbb{K}^2$ are classified by

- ▶ The integral affine structure on the base.
- ▶ An element $[s] \in H^2(\Lambda)$.
- ▶

$$H^2(\Lambda) = \begin{cases} \frac{\mathbb{Z}}{2\mathbb{Z}} \oplus \mathbb{Z}, & \text{for family 1) and family 2) with } n \equiv 0 \pmod{2}. \\ \mathbb{Z}, & \text{otherwise.} \end{cases}$$

Classification of Integral Affine Structures on $\mathbb{K}^2$ - $\pi_1(K^2) = \langle a, b \mid aba = b \rangle$

Let $A_1 = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$, $A_2 = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ and $B_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $B_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $B_3 = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$

Theorem (Sepe, 2010)

The possible generators for the affine holonomy group are

- 1 $(I, x\mathbf{e}_2)$ and $(B_1, y\mathbf{e}_1)$ with $x, y > 0$.
- 2 $(A_1, x\mathbf{e}_2)$ and $(B_1, y\mathbf{e}_1 + x\mathbf{e}_2)$ with $x, y > 0$.
- 3 $(I, x(\mathbf{e}_1 - \mathbf{e}_2))$ and $(B_2, y\mathbf{e}_2)$ with $x, y > 0$.
- 4 $(A_2, x\mathbf{e}_2)$ and $(B_3, y\mathbf{e}_1 + \frac{n-1}{n}x\mathbf{e}_2)$ with $2y \neq \frac{n-1}{n}x$ and $x > 0$.

Questions?

Questions?

References I



Duistermaat, J. J. (1980). On global action-angle coordinates. *Communications on Pure and Applied Mathematics*, 33(6), 687–706.

<https://doi.org/https://doi.org/10.1002/cpa.3160330602>



Ehresmann, C. (1936). Sur les espaces localement homogènes. *Annales de l'École Normale Supérieure*, 53, 317–333.



Ehresmann, C. (1951). Les connexions infinitésimales dans un espace fibré différentiable. *Colloque de topologie (espaces fibrés), Bruxelles, 1950*, 29–55.



Fried, D., Goldman, W., & Hirsch, M. (1981). Affine manifolds with nilpotent holonomy. *Commentarii Mathematici Helvetici*, 56, 487–523.

<https://doi.org/10.1007/BF02566225>



Mishachev, K. (1996). The classification of lagrangian bundles over surfaces. *Differential Geometry and its Applications*, 6(4), 301–320.

[https://doi.org/https://doi.org/10.1016/S0926-2245\(96\)00024-1](https://doi.org/https://doi.org/10.1016/S0926-2245(96)00024-1)

References II



Sepe, D. (2010). Classification of lagrangian fibrations over a klein bottle. *Geometriae Dedicata*, 149(1), 347–362. <https://doi.org/10.1007/s10711-010-9485-8>